# **JACOB** PRUD'HOMME

**Recent Graduate with 3.5 Years of Experience**


jacobprudhomme
jacobprudhom.me
Zürich, Switzerland
jacob-prudhomme
contact@jacobprudhom.me
0000-0002-6108-6587

## SELECTED WORK EXPERIENCE

### Cryptography Engineering Intern — **evolutionQ**

Mar 2025 – Aug 2025 | Aachen, Germany

- Rearchitected cryptography backend for quantum-safe symmetric key distribution (SKI) protocol written in Rust, allowing it to interface with PKCS#11 devices such as HSMs, enabling support for hardened security options
- Started extending cryptography backend to support new post-quantum capabilities added in PKCS#11 3.2
- Analysed flow of secrets throughout implementation and refactored it to ensure sensitive key material stays within the HSM boundary until absolutely necessary

### Junior Software Engineer — **Mission Control Space Services**

Feb 2022 – Sep 2024 | Ottawa, Canada

- Laid groundwork for embedded software project to test deep learning capabilities on constrained space hardware, using ESA's OPS-SAT "orbital laboratory" satellite, which runs on an Intel Cyclone V system-on-a-chip
- Managed framework for collecting data for satellite position determination, providing a robust dataset for deep learning
- Redesigned data model of open-source web app for defining Interface Control Document models with ability to export to code or data, streamlining integration with external codebases, space flight software and ground control systems
- Integrated flagship mission control software with various robotics backends to meet time-critical mission/project requirements for a variety of stakeholders

### Software Developer (Cybersecurity) — **Crypto4A**

Sep 2020 – Dec 2020 | Ottawa, Canada

- Led initiative to design and develop Vue.js visualization front-end for displaying the current state of an M-of-N quorum access control system, along with companion app in Kotlin
- Helped attain FIPS 140-2 security compliance by building parts of ACVP cryptographic testing framework in Java
- Improved core services, written in C, running on flagship quantum-safe "datacentre-in-a-box" appliance

### Software Engineering Intern — **SigOpt (acquired by Intel)**

Jan 2020 – Mar 2020 | San Francisco, USA

- Architected and implemented team collaboration capabilities for new tool to manage workflows for developing machine learning models, from PostgreSQL database design, to back-end API design, to UX in React
- Continuously improved React frontend and Python backend for flagship hyperparameter optimization product

## EDUCATION

### MSc in Cyber Security

**EPFL and ETH Zürich (joint degree)**

Sep 2022 – Mar 2026

### Bachelor of Computer Science

Minor in Combinatorics & Optimization

**University of Waterloo**

Sep 2016 – Aug 2021

## MORE EXPERIENCE

### Semester Research Project Student

**LASEC – Cryptography and Security Lab**

2023 – 4 months | EPFL

### Undergraduate Research Assistant

**Programming Languages Group**

2021 – 4 months | Uni. of Waterloo

### Secure Software Developer

**ESCRYPT**

2019 – 4 months | Waterloo, Canada

### Full-Stack Developer

**360insights**

2018 – 8 months | Whitby, Canada

### Web Applications Developer

**Blindside Networks**

2017 – 4 months | Ottawa, Canada

## PUBLICATIONS

### Conference Proceedings

- L. Chavier et al., "Deploying artificial intelligence capabilities by hybridizing a neural network on a satellite," in *ASCEND 2023*. doi: 10.2514/6.2023-4718 eprint: https://arc.aiaa.org/doi/pdf/10.2514/6.2023-4718.

## LANGUAGES

English (Native) | French (Native)

## SELECTED ACADEMIC EXPERIENCE

### Gotta Study 'Em All: An efficiency and security analysis of the POKÉ family of PKEs from HD isogenies

*Master Thesis*

**IBM Research Zürich and Eötvös Loránd University (ELTE)**

Supervisors: Andrea Basso (IBM Research), Péter Kutas (ELTE)

Sep 2025 – Mar 2026 | Zürich, CH and Budapest, HU

- Developed cryptographic-grade reference implementations in Rust for POKÉ and INKE, recent post-quantum PKE schemes based on higher-dimensional isogenies
- Identified and explored improvements at the protocol and implementation levels to optimize performance by ~1.5-3.0x over previous proof-of-concept implementations in C
- Undertook cryptanalysis of the schemes using novel insights from publication *Leveled Isogeny Problems with Hints* [Das et al., PKC 2026], partially developing two attacks against INKE

### EPFL-ETH Summer School on Lattice Cryptography

**EPFL and ETH Zürich**

Jul 2025 | EPFL, Switzerland

- Attended talks on the theoretical foundations of lattice cryptography as well as on the specific workings of ML-KEM and ML-DSA (given by the researchers involved in the standards)
- Covered advanced topics: optimized implementations, proof systems based on lattices, fully homomorphic encryption, functional encryption, and oblivious pseudorandom functions

### ACME Client

*Course Project*

**Network Security Group**

Fall Semester 2024 | ETH Zürich, Switzerland

- Developed understanding of standards and interpreted specifications spread over multiple RFCs to implement client for automated certificate management protocol in Rust
- Used Tokio runtime for async operation and set up with Rust-idiomatic argument parsing, logging and error handling

### SecretScroll: A privacy-preserving navigation app

*Course Project*

**SPRING – Security and Privacy Engineering Lab**

Spring Semester 2024 | EPFL, Switzerland

- Integrated technologies such as attribute-based credentials and applied techniques like spatial obfuscation and dummy locations to achieve location privacy, while still balancing utility
- Created machine learning model to run network fingerprinting attack using metadata collected with Wireshark and tcpdump

### Implementation of pSIDH

*Semester Research Project*

**LASEC – Cryptography and Security Lab**

Supervisor: Laurane Marco

Spring Semester 2023 | EPFL, Switzerland

- Created reference implementation in SAGE of pSIDH, a post-quantum key exchange based on the Deuring correspondence

## OPEN-SOURCE PROJECTS

- rust-cryptoki: Idiomatic Rust PKCS#11 bindings
- isogeny_rs: Isogeny cryptography library for Rust
- poke-rs: Reference implementation in Rust for isogeny-based PKE POKÉ (as part of master thesis)
- pSIDH: Reference implementation in SAGE for isogeny-based key exchange pSIDH (as part of semester research project)

## SKILLS

**Languages**: C, Javascript / Typescript, Rust, Python / SAGE, HTML, CSS, PHP, Scala, Kotlin, Haskell

**Tools**: Bash, Git, GitHub Actions

**Cloud / Infrastructure**: AWS, Azure, Docker, Kubernetes, Puppet, Istio, Hashicorp Vault

**Frontend**: React, Vue.js, Svelte

**Backend**: Express.js, Flask, Axum

**Databases**: PostgreSQL, MongoDB, Redis

**Runtimes**: Node.js, Tokio

**Concepts**: Microservices architecture, REST APIs, gRPC / Protobuf